



Soknarith (Soki) Sem

☎ (401) 952-6819 | ✉ seminaia401@gmail.com | 🌐 [Seminaia](#) | in [soknarith-sem](#) | 📍 Worcester, MA

SUMMARY

Ph.D. candidate in materials science and engineering with a background in chemical engineering and applied mathematics. Experienced in atomistic modeling, thermodynamics, and transport phenomena, with 2 years of hands-on experience in laboratory and manufacturing. Proficient in scientific computing tools including Python, MATLAB/Simulink, ASPEN Plus Dynamics, FreeCAD, and R for data analysis.

WORK EXPERIENCE

Research Assistant / Graduate Teaching Assistant

May 2024 to Present

Worcester Polytechnic Institute – Worcester, MA

- Quantified defect energetics and transport behavior in oxide perovskites using DFT and thermodynamic modeling.
- Analyzed structure–property–process relationships across temperature and chemical potential regimes.
- Assisted in teaching undergraduate and graduate materials engineering courses by grading assignments and leading homework help sessions.

Chemical Process Engineer

May 2023 to May 2024

Knowles Precision Devices – New Bedford, MA

- Led the installation of new impregnation systems for high-voltage capacitors.
- Read P&ID and process flow diagrams to ensure proper equipment setup and process flow.
- Applied Lean/Six-Sigma/5S principles and statistical process controls in order to operate within KPI targets.
- Authored SOPs, technical documentation, safety protocols for material handling and process scale-up.
- Mentored technicians and operators on process procedures, safety protocols, and quality standards.
- Performed root cause analysis on batch failures, including CAPA implementation and process improvement suggestions.
- Modeled heater behavior in order to quantify the length of time required to reach target temperatures and pressures.

R&D Technician

Mar 2023 to May 2023

Factorial Energy – Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments.
- Reduced material waste by 20% through inventory system redesign.

Downstream Purification Engineer

Jul 2022 to Mar 2023

Eurofins Scientific – Framingham, MA

- Performed lab-scale separation and purification of proteins using AKTA Avant chromatography systems achieving 95–99% purity.
- Applied DOE methods to optimize buffer conditions and flow rate for maximum yield and purity at lab-scale.
- Prepared buffers, reagents, and packed chromatography columns for lab-scale purification runs.
- Conducted lab-scale TFF membrane filtration, UF/DF, and AKTA chromatography resin life-time studies to validate CIP protocols.
- Characterized purified proteins using SDS-PAGE, HPLC/MS, GC/MS, UV-Vis, and related analytical techniques.

EDUCATION

Ph.D. Materials Science & Engineering

May 2028

Worcester Polytechnic Institute – Worcester, MA

M.S. Materials Science & Engineering

May 2026

Worcester Polytechnic Institute – Worcester, MA

B.S. Chemical Engineering

May 2022

University of Rhode Island – Kingston, RI

B.S. Applied Mathematics

May 2022

University of Rhode Island – Kingston, RI

Minors: Chemistry, Physics

TECHNICAL SKILLS

Atomistic Simulation:

DFT (VASP & GPAW), Molecular Dynamics (LAMMPS)

Scientific Computing:

Python, MATLAB/Simulink, ASPEN Plus Dynamics, Linux, LaTeX, R, FreeCAD

Laboratory / Analytical:

Glovebox, clean-room, HPLC, AKTA FPLC, TFF, UF/DF, SEM/XRD, DSC/TGA, UV-Vis, Osmometry, HPLC/MS, GC/MS, pH meter, SDS-PAGE, KF coulometric titration

Process / Manufacturing:

SPC, DOE, Lean/Six-Sigma, Process optimization, SOP development, Tech Transfer (MSAT), cGMP